(FP2) 5. Complex numbers 作业

NO. 1

Find all the roots of the equation $(w+1)^6 = 1$, giving your answers in the form $x+iy$ where x and y are real and exact. [4]

NO. 2

a. Use de Moivre's theorem to show that $\sin^6\theta = -\frac{1}{32}(\cos 6\theta - 6\cos 4\theta + 15\cos 2\theta - 10)$. [6]

It is given that $\cos^6\theta = \frac{1}{32}(\cos 6\theta + 6\cos 4\theta + 15\cos 2\theta + 10)$.

b. Find the exact value of $\int_0^{\frac{1}{3}\pi} \left(\cos^6(\frac{1}{4}x) + \sin^6(\frac{1}{4}x)\right) \mathrm{d}x$. [4]

c. Express each root of the equation $16c^6 + 16(1 - c^2)^3 - 13 = 0$ in the form $\cos k\pi$, where k is a rational number. [5]

NO. 3

a. Find the roots of the equation $z^3 = -1 - i$, giving your answers in the form $re^{i\theta}$, where $r > 0$ and $0 \leq \theta \leq 2\pi$. [5]

Let $w = z_1^{3k} + z_2^{3k} + z_3^{3k}$, where k is a positive integer and z_1, z_2, z_3 are the roots of $z^3 = -1 - i$.

b. Express w in the form $Re^{i\alpha}$, where $R > 0$, giving R and α in terms of k. [3]

NO. 4

i. Use de Moivre's theorem to show that

$$\cos 4\theta = \cos^4 \theta - 6 \cos^2 \theta \sin^2 \theta + \sin^4 \theta. \quad [3]$$

ii. Hence find all the roots of the equation

$$x^4 - 6x^2 + 1 = 0$$

in the form $\tan q\pi$, where q is a positive rational number. [5]

NO. 5

i. Use de Moivre's theorem to show that

$$\sin 8\theta = 8 \sin\theta \cos\theta(1 - 10 \sin^2\theta + 24 \sin^4\theta - 16 \sin^6\theta).$$ [6]

ii. Use the equation $\frac{\sin 8\theta}{\sin 2\theta} = 0$ to find the roots of

$$16x^6 - 24x^4 + 10x^2 - 1 = 0$$

in the form $\sin k\pi$, where k is rational. [4]

NO. 6

i. Use de Moivre's theorem to prove that

$$\tan 4\theta = \frac{4\tan\theta - 4\tan^3\theta}{1 - 6\tan^2\theta + \tan^4\theta}. \quad [5]$$

ii. Hence find the solutions of the equation

$$t^4 - 4t^3 - 6t^2 + 4t + 1 = 0,$$

giving your answers in the form tan $k\pi$, where k is a rational number. [5]

NO. 7

i. Use de Moivre's theorem to show that

$$\sin 5\theta = 5\sin\theta - 20\sin^3\theta + 16\sin^5\theta. \quad [5]$$

ii. Hence explain why the roots of the equation $16x^4 - 20x^2 + 5 = 0$ are $x = \pm\sin\frac{1}{5}\pi$ and $x = \pm\sin\frac{2}{5}\pi$. [3]

iii. Without using a calculator, find the exact values of

$$\sin\tfrac{1}{5}\pi\sin\tfrac{2}{5}\pi\sin\tfrac{3}{5}\pi\sin\tfrac{4}{5}\pi \quad \text{and} \quad \sin^2\left(\tfrac{1}{5}\pi\right) + \sin^2\left(\tfrac{2}{5}\pi\right). \quad [4]$$

NO. 8

a. Use de Moivre's theorem to show that $\cot 6\theta = \dfrac{\cot^6\theta - 15\cot^4\theta + 15\cot^2\theta - 1}{6\cot^5\theta - 20\cot^3\theta + 6\cot\theta}$. [6]

b. Hence obtain the roots of the equation

$$x^6 - 6x^5 - 15x^4 + 20x^3 + 15x^2 - 6x - 1 = 0$$

in the form $\cot(q\pi)$, where q is a rational number. [4]

NO. 9

a. Show that $\sum_{r=1}^{n} z^{4r} = \frac{z^{4n+2} - z^2}{z^2 - z^{-2}}$, for $z^2 \neq z^{-2}$. [2]

b. By letting $z = \cos\theta + i\sin\theta$, show that, if $\sin 2\theta \neq 0$,

$$\sum_{r=1}^{n} \sin(4r\theta) = \frac{\cos 2\theta - \cos(4n+2)\theta}{2\sin 2\theta}.$$ [5]

NO. 10

By considering the binomial expansions of $\left(z+\frac{1}{z}\right)^4$ and $\left(z-\frac{1}{z}\right)^4$, where $z = \cos\theta + i\sin\theta$, use de Moivre's theorem to show that

$$\cot^4\theta = \frac{\cos 4\theta + a\cos 2\theta + b}{\cos 4\theta - a\cos 2\theta + b},$$

where a and b are integers to be determined. [7]

NO. 11

a. State the sum of the series $1 + w + w^2 + w^3 + \ldots + w^{n-1}$, for $w \neq 1$. [1]

b. Show that $(1 + i\tan\theta)^k = \sec^k\theta(\cos k\theta + i\sin k\theta)$, where θ is not an integer multiple of $\frac{1}{2}\pi$. [2]

c. By considering $\sum_{k=0}^{n-1}(1+i\tan\theta)^k$, show that

$$\sum_{k=0}^{n-1}\sec^k\theta\sin k\theta = \cot\theta(1-\sec^n\theta\cos n\theta),$$

provided θ is not an integer multiple of $\frac{1}{2}\pi$. [5]

d. Hence find $\sum_{k=0}^{6m-1}2^k\sin\left(\frac{1}{3}k\pi\right)$ in terms of m. [2]

NO. 12

a. Find a and b such that $z^8 - iz^5 - z^3 + i = (z^5 - a)(z^3 - b)$. [1]

b. Hence find the roots of

$$z^8 - iz^5 - z^3 + i = 0,$$

giving your answers in the form $re^{i\theta}$, where $r > 0$ and $0 \leqslant \theta < 2\pi$. [6]